

Notes: Odean (JF, 1998)

Are Investors Reluctant to Realize Their Losses?

Contents

1	Big picture	3
2	Data and measurement (key objects)	4
2.1	Brokerage data and sample	4
2.2	Reconstructing investor portfolios	5
2.3	Reference points and gain/loss classification	5
2.4	Realized and paper gains/losses	6
2.5	Key descriptive objects	6
2.6	Unobservables and limitations	6
3	Models and empirical specifications (all equations + interpretation)	7
3.1	Prospect-theory intuition	7
3.2	Proportion of Gains Realized	7
3.3	Proportion of Losses Realized	7
3.4	Main disposition-effect hypothesis	8
3.5	Tax-loss-selling hypothesis	8
3.6	Difference-in-proportions test	8
3.7	Account-level robustness check	9
3.8	Rebalancing controls	9
3.9	Ex post performance test	9
3.10	Bootstrap performance inference	10
3.11	Additional purchases test	10
4	Identification and instruments (what makes the estimates credible)	10
4.1	What fails in the simple approach	10
4.2	Main source of identification	10

4.3	Why the design is credible	11
4.4	Alternative explanations addressed	11
4.5	What the paper cannot fully separate	11
5	Tables and figures	12
5.1	Figure 1: prospect theory value function	12
5.2	Table I: PGR and PLR for the full data set	12
5.3	Figure 2: monthly ratio of PGR to PLR	13
5.4	Table II: partitions by period and trading activity	13
5.5	Table III: average returns on realized and paper positions	14
5.6	Tables IV and V: rebalancing checks	14
5.7	Table VI: ex post returns	15
5.8	Table VII: price and return partitions	15
5.9	Additional-purchases evidence	16
5.10	Discussion calculation: economic cost	16
6	Short summary	16
7	Conclusion	17
7.1	Core conclusion	17
7.2	Economic intuition	17
7.3	Takeaway	17

1 Big picture

- **Question.** Do individual investors exhibit the **disposition effect**: the tendency to sell winning investments too soon and hold losing investments too long?
- **Setting.** The paper studies trading records from **10,000 accounts** at a large U.S. discount brokerage house from **1987 through 1993**. The empirical object is the rate at which investors realize gains versus the rate at which they realize losses.
- **Core behavioral idea.** If investors evaluate stocks relative to a reference point, usually the purchase price, then a stock trading above that reference point sits in the gain region, where the investor is relatively risk averse. A stock trading below the reference point sits in the loss region, where the investor may become risk seeking. This can make investors eager to lock in gains and reluctant to lock in losses.
- **Core empirical challenge.** It is not enough to count whether investors sell more winners than losers. In a rising market, investors mechanically hold more winners. The right comparison must normalize by **opportunities**: how often investors sell winners relative to all winners they could have sold, and how often they sell losers relative to all losers they could have sold.
- **Main contribution.** The paper builds two realization-rate measures:
 - the **Proportion of Gains Realized** (PGR);
 - the **Proportion of Losses Realized** (PLR).

The disposition effect predicts $\text{PGR} > \text{PLR}$ outside tax-loss-selling periods.

- **Competing explanations.** The paper does not simply document selling winners and holding losers. It asks whether the pattern can instead be explained by:
 - portfolio rebalancing;
 - lower expected future returns on winners;
 - mean-reversion beliefs;
 - higher transaction costs for low-priced loser stocks;
 - year-end tax-loss selling.
- **Why it matters.** The result is one of the classic field-data tests of behavioral finance. It shows that a bias documented in experiments and theory appears strongly in real investor trading records, with direct consequences for after-tax returns.
- **Main empirical question.** Conditional on having a chance to sell winners or losers, do individual investors realize gains at a higher rate than losses?
- **Main empirical results.**
 - For the full year, $\text{PGR} = 0.148$ and $\text{PLR} = 0.098$. A winner is therefore more than **50% more likely** to be sold than a loser.
 - From January through November, the disposition effect is stronger: $\text{PGR} = 0.152$ and $\text{PLR} = 0.094$.
 - In December the pattern reverses: $\text{PGR} = 0.108$ and $\text{PLR} = 0.128$, consistent with tax-loss selling.

- The ratio PGR/PLR declines from about **2.1 in January** to about **0.85 in December**.
- The result is robust to alternative independence assumptions, share-based and dollar-based measures, partitions by time period, and partitions by frequent versus infrequent traders.
- It is not explained by rebalancing: the result remains when using only sales of an entire position and when excluding sales followed by new purchases.
- It is not explained by ex post performance: winners investors sell continue to outperform paper losers investors keep by about **3.41 percentage points** over the next 252 trading days.
- It is not explained by low-price transaction costs: the disposition effect remains within price and return-magnitude bins.
- Investors are also more likely to buy additional shares of losers than winners: $PLPA = 0.135$ versus $PGPA = 0.094$.
- **Economic takeaway.** Individual investors appear reluctant to admit losses. They sell winners too readily, hold losers too long, and therefore give up both tax benefits and subsequent performance. The paper's central message is that reference-dependent preferences shape real trading behavior.

2 Data and measurement (key objects)

2.1 Brokerage data and sample

- **Source.** A nationwide discount brokerage house provides trading records.
- **Account sample.** From all accounts active in 1987, meaning accounts with at least one transaction, the brokerage randomly selects **10,000 customer accounts**.
- **Sample period.** January 1987 through December 1993.
- **Raw files.** The brokerage provides:
 - a trades file;
 - a security-number-to-CUSIP file;
 - a monthly positions file.

The main analysis uses the trades file and the CUSIP translation file.

- **Trades file.** The trades file contains **162,948 records**. Each record includes:
 - account identifier;
 - trade date;
 - internal security number;
 - buy/sell indicator;
 - quantity;
 - commission;
 - principal amount.

- **Aggregation.** Multiple buys or sells of the same stock in the same account on the same day are aggregated.
- **CRSP match.** The paper limits the empirical analysis to stocks with CRSP price, return, split, and dividend information. In the final CRSP-linked stock-trading sample, **6,380 accounts** trade CRSP stocks, generating **97,483 transactions**.
- **Account types.** The data do not identify whether an account is taxable or tax-exempt. At the beginning of the period, about **20%** of brokerage accounts are IRA or Keogh accounts, and these accounts account for about **17.5%** of trades. Their inclusion makes tax effects harder, not easier, to detect.
- **Survivorship issue.** Accounts closed during the sample are not replaced. This can bias the account sample toward more successful investors, but it is not an obvious reason for the specific gain-versus-loss realization pattern.

2.2 Reconstructing investor portfolios

- The paper reconstructs each account's portfolio by moving chronologically through the account's trading records.
- The reconstructed portfolio includes only positions for which the purchase date and purchase price are known from the sample period.
- Stocks purchased before January 1987 may be held in the account but cannot be assigned a purchase price in the data. Those pre-sample positions are therefore not central to the reconstructed portfolio.
- The reconstructed portfolio is only part of each investor's total wealth portfolio. Investors may hold other brokerage accounts, mutual funds, retirement accounts, or other assets outside the sample.
- This partial-portfolio limitation matters, but the key test compares realized winners to unrealized winners and realized losers to unrealized losers within the same reconstructed trading records.

2.3 Reference points and gain/loss classification

- The paper assumes the main reference point is the stock's **average purchase price**.
- The author notes that other reference points are possible, such as:
 - first purchase price;
 - most recent purchase price;
 - highest purchase price;
 - a path-dependent aspiration level.

The empirical findings are similar under alternative purchase-price reference points.

- For realized sales, the sale price is compared to the average purchase price.
- Commissions are included in the purchase price and deducted from the sale price because capital-gains taxes depend on transaction costs.

- Dividends are not included when determining capital gains or losses because they do not affect capital gain tax treatment.
- All gains and losses are adjusted for stock splits.

2.4 Realized and paper gains/losses

- On each day a sale occurs in an account with at least two stocks, the paper classifies each sold stock as a realized gain or a realized loss.
- For each stock held at the beginning of that day but not sold:
 - if both the daily high and daily low are above average purchase price, it is counted as a paper gain;
 - if both the daily high and daily low are below average purchase price, it is counted as a paper loss;
 - if the purchase price lies between the daily high and low, it is counted as neither.
- On days when no sale occurs in an account, no realized or paper gain/loss observations are counted.
- This design makes the denominator an **opportunity set**: gains and losses that could have been realized on days when the investor was demonstrably willing to trade.

2.5 Key descriptive objects

- **Realized gain**: a sold stock whose sale value exceeds its purchase-cost basis.
- **Realized loss**: a sold stock whose sale value is below its purchase-cost basis.
- **Paper gain**: an unsold stock that could have been sold for a gain on a sale day.
- **Paper loss**: an unsold stock that could have been sold for a loss on a sale day.
- **Proportion realized**: the rate at which investors convert possible gains or possible losses into actual sales.

2.6 Unobservables and limitations

- The data do not reveal investors' expectations, risk tolerance, tax brackets, or true reference points.
- The data do not reveal whether an account is taxable or tax-exempt.
- The data do not observe investors' total portfolios outside this brokerage.
- The statistical tests initially treat realized gains, realized losses, paper gains, and paper losses as independent observations. The paper acknowledges that this is not literally true, and therefore reports account-level robustness checks.
- The paper can strongly reject several rational explanations, but it does not fully separate prospect theory from a mistaken belief in mean reversion. Both can imply selling winners and holding losers.

3 Models and empirical specifications (all equations + interpretation)

3.1 Prospect-theory intuition

A common way to formalize the prospect-theory value function is:

$$v(x) = \begin{cases} x^\alpha, & x \geq 0, \\ -\lambda(-x)^\beta, & x < 0, \end{cases} \quad \lambda > 1. \quad (1)$$

Here x is a gain or loss relative to a reference point. In the paper's empirical application, the reference point is proxied by purchase price.

Interpretation. The value function is concave over gains and convex over losses. This means investors are relatively risk averse after a gain but relatively risk seeking after a loss. If a stock has risen above purchase price, the investor is more willing to sell. If a stock has fallen below purchase price, the investor is more willing to gamble by continuing to hold.

3.2 Proportion of Gains Realized

Let:

- RG = number of realized gains;
- PG = number of paper gains.

The Proportion of Gains Realized is:

$$\text{PGR} = \frac{\text{RG}}{\text{RG} + \text{PG}}. \quad (2)$$

Interpretation. This is the probability-like rate at which investors sell winning positions, conditional on having winning positions that could be sold on days when they sell something.

3.3 Proportion of Losses Realized

Let:

- RL = number of realized losses;
- PL = number of paper losses.

The Proportion of Losses Realized is:

$$\text{PLR} = \frac{\text{RL}}{\text{RL} + \text{PL}}. \quad (3)$$

Interpretation. This is the corresponding realization rate for losing positions. The disposition effect predicts that investors realize gains faster than losses, so $\text{PGR} > \text{PLR}$.

3.4 Main disposition-effect hypothesis

The first hypothesis is:

$$H_1 : \text{PGR} > \text{PLR}. \quad (4)$$

The null is:

$$H_0 : \text{PGR} \leq \text{PLR}. \quad (5)$$

Interpretation. If investors are indifferent between selling winners and losers, after conditioning on opportunities, the two realization rates should be similar. A large positive gap $\text{PGR} - \text{PLR}$ means investors are more eager to lock in gains than losses.

3.5 Tax-loss-selling hypothesis

The second hypothesis asks whether December differs from the rest of the year:

$$H_2 : (\text{PLR}_D - \text{PGR}_D) > (\text{PLR}_{J-N} - \text{PGR}_{J-N}), \quad (6)$$

where D denotes December and $J-N$ denotes January through November.

Interpretation. Taxable investors should realize losses before the tax year closes. Thus, even if they usually hold losers, December may produce unusually high loss realization.

3.6 Difference-in-proportions test

The standard error used for the difference between the two proportions is:

$$SE(\text{PGR} - \text{PLR}) = \sqrt{\frac{\text{PGR}(1 - \text{PGR})}{n_{rg} + n_{pg}} + \frac{\text{PLR}(1 - \text{PLR})}{n_{rl} + n_{pl}}}, \quad (7)$$

where n_{rg} , n_{pg} , n_{rl} , and n_{pl} are the counts of realized gains, paper gains, realized losses, and paper losses.

A corresponding test statistic is:

$$t = \frac{\text{PGR} - \text{PLR}}{SE(\text{PGR} - \text{PLR})}. \quad (8)$$

Interpretation. The paper's baseline tests treat each realized or paper gain/loss as an observation. The author recognizes that this may overstate precision because repeated decisions by the same investor or common information across investors may create dependence.

3.7 Account-level robustness check

For each account i , define:

$$d_i = \text{PGR}_i - \text{PLR}_i. \quad (9)$$

The robustness test asks whether:

$$E[d_i] > 0. \quad (10)$$

Interpretation. This test assumes independence only across accounts, not across trades. It weights accounts equally rather than giving larger weight to more active accounts. The result remains very strong: the average account-level PGR is about 0.57, the average account-level PLR is about 0.36, and the average gap is about 0.21.

3.8 Rebalancing controls

The paper uses two restrictions to test whether the PGR–PLR gap is simply portfolio rebalancing:

1. count only cases where the investor sells the **entire position** in a stock;
2. count only sales for which the investor makes **no new purchase** into the portfolio on the sale day or during the following three weeks.

Interpretation. If investors sell winners only to rebalance, the disposition effect should weaken substantially after these restrictions. It does not. The preference for selling winners remains.

3.9 Ex post performance test

For a stock j over horizon h , define a buy-and-hold excess return relative to the CRSP value-weighted index:

$$\text{BHAR}_{j,h} = \prod_{\tau=1}^h (1 + r_{j,\tau}) - \prod_{\tau=1}^h (1 + r_{m,\tau}). \quad (11)$$

The paper compares:

$$E[\text{BHAR}_h \mid \text{winning stock sold}] - E[\text{BHAR}_h \mid \text{paper loser kept}]. \quad (12)$$

Interpretation. If investors rationally sell winners and hold losers because losers are expected to rebound, paper losers should subsequently outperform winners sold. The paper finds the opposite: winners sold subsequently outperform paper losers.

3.10 Bootstrap performance inference

The paper constructs empirical p-values by drawing replacement stocks matched on size decile and book-to-market quintile, then recomputing the excess-return differences many times.

Interpretation. This addresses the concern that the subsequent performance comparison is driven by the types of stocks in the winning and losing groups rather than by investors' sale decisions.

3.11 Additional purchases test

To distinguish the transaction-cost story from behavioral stories, the paper studies whether investors buy additional shares of existing winners or existing losers.

Define:

$$\text{PGPA} = \frac{\text{gains purchased again}}{\text{gains purchased again} + \text{gains potentially purchased again}}, \quad (13)$$

and

$$\text{PLPA} = \frac{\text{losses purchased again}}{\text{losses purchased again} + \text{losses potentially purchased again}}. \quad (14)$$

Interpretation. If investors avoid losers merely because loser stocks have high transaction costs, they should also be reluctant to buy additional loser shares. Instead, the paper finds $\text{PLPA} = 0.135$ and $\text{PGPA} = 0.094$. Investors are more likely to double down on losers than on winners, which supports a behavioral explanation over a pure transaction-cost explanation.

4 Identification and instruments (what makes the estimates credible)

4.1 What fails in the simple approach

- Simply counting realized gains and realized losses is not enough.
- If the stock market rises, investors will naturally have more winners than losers. More winner sales may simply reflect more winner opportunities.
- The paper's key empirical move is therefore to compare realized gains to **all available gains** and realized losses to **all available losses**.

4.2 Main source of identification

- The identifying comparison is between two conditional realization rates:

$$\text{PGR} - \text{PLR}. \quad (15)$$

- The opportunity-set denominator controls for how many winning or losing positions the investor could have sold.

- The comparison is made only on days when the investor sells something, which focuses on moments when liquidity, attention, and willingness to trade are present.
- The gain/loss classification uses the purchase price as the reference point, giving a direct test of reference-dependent selling behavior.

4.3 Why the design is credible

- **Large account-level trading data.** The study uses actual trading records rather than survey responses or lab choices.
- **Opportunity normalization.** PGR and PLR normalize by possible sales, not just actual sales.
- **Discount brokerage setting.** Because the accounts are from a discount brokerage, the results are less likely to reflect retail brokers pushing clients to sell particular stocks.
- **Robust to account-level dependence.** The result survives a test that assumes independence only across accounts.
- **Robust to share and dollar weighting.** The result is not an artifact of counting small gains and large losses differently.
- **Robust across subgroups.** The result appears in both earlier and later sample periods and among both frequent and infrequent traders.

4.4 Alternative explanations addressed

- **Rebalancing.** If investors sell winners to rebalance, the effect should shrink when looking only at full-position sales or sales not followed by purchases. It does not.
- **Rational expectations about future returns.** If held losers are expected to outperform sold winners, then subsequent returns should justify the choices. They do not: sold winners outperform paper losers.
- **Low-price transaction costs.** Loser stocks are often lower priced, and low-priced stocks can have higher trading costs. The paper partitions by price and return magnitude and still finds the disposition effect in nearly all cells.
- **Tax-loss selling.** The December reversal is not a threat to the disposition effect; it is a separate prediction. Investors usually realize winners more readily, but tax motives become more visible in December.
- **Common information shocks.** Multiple investors may sell the same stock around the same time because of shared news. The account-level robustness checks and high statistical magnitudes reduce this concern.

4.5 What the paper cannot fully separate

- The paper's evidence is consistent with prospect theory.
- It is also consistent with a mistaken belief that losing stocks will mean revert.

- The subsequent-return results show that such mean-reversion beliefs are not justified on average, but they do not prove that investors consciously use prospect-theory preferences rather than mistaken expectations.
- The paper also cannot fully observe taxes, outside wealth, or exact investor-specific reference points.

5 Tables and figures

5.1 Figure 1: prospect theory value function

Read:

- The value function is S-shaped around a reference point.
- It is concave for gains and convex for losses.
- It is steeper for losses than gains.
- This creates the core intuition: investors may be risk averse with winners and risk seeking with losers.

Takeaway. If purchase price is the reference point, investors are psychologically more willing to close winning positions than losing positions.

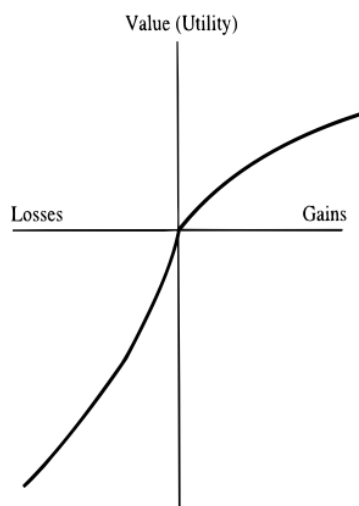


Figure 1. Prospect theory value function.

Table I
PGR and PLR for the Entire Data Set

This table compares the aggregate Proportion of Gains Realized (PGR) to the aggregate Proportion of Losses Realized (PLR), where PGR is the number of realized gains divided by the number of realized gains plus the number of paper (unrealized) gains, and PLR is the number of realized losses divided by the number of realized losses plus the number of paper (unrealized) losses. Realized gains, paper gains, losses, and paper losses are aggregated over time (1987–1993) and across all accounts in the data set. PGR and PLR are reported for the entire year, for December only, and for January through November. For the entire year there are 13,883 realized gains, 79,658 paper gains, 11,930 realized losses, and 110,348 paper losses. For December there are 886 realized gains, 7,131 paper gains, 1,555 realized losses, and 10,604 paper losses. The *t*-statistics test the null hypotheses that the differences in proportions are equal to zero assuming that all realized gains, paper gains, realized losses, and paper losses result from independent decisions.

	Entire Year	December	Jan.–Nov.
PLR	0.098	0.128	0.094
PGR	0.148	0.108	0.152
Difference in proportions	–0.050	0.020	–0.058
<i>t</i> -statistic	–35	4.3	–38

5.2 Table I: PGR and PLR for the full data set

	Entire year	December	Jan.–Nov.
PLR	0.098	0.128	0.094
PGR	0.148	0.108	0.152
PGR – PLR	0.050	–0.020	0.058

Read:

- For the entire year, investors realize gains at a much higher rate than losses.
- From January through November, the gap is even larger.
- In December, the sign reverses because loss realization becomes more common.

Takeaway. The baseline table is the central evidence: investors strongly prefer realizing winners, except when tax-loss selling becomes salient at year-end.

5.3 Figure 2: monthly ratio of PGR to PLR

Read:

- The ratio PGR/PLR is about 2.1 in January.
- The ratio remains above 1 for January through November.
- The ratio declines through the year and reaches about 0.85 in December.

Takeaway. The disposition effect dominates during most of the year, while tax-loss selling becomes strongest near the tax deadline.

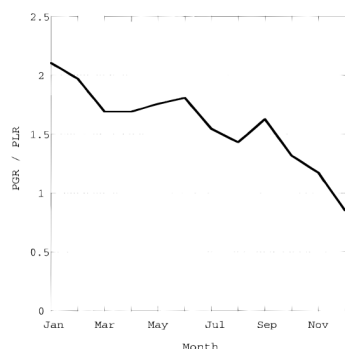


Figure 2. Ratio of the Proportion of Gains Realized (PGR) to the Proportion of Losses Realized (PLR) for each month. PGR is the number of realized gains divided by the number of realized gains plus the number of paper (unrealized) gains, and PLR is the number of realized losses divided by the number of realized losses plus the number of paper (unrealized) losses. Realized gains, paper gains, losses, and paper losses are aggregated over time (1987–1993) and across all accounts in the data set.

Table II
PGR and PLR Partitioned by Period and Trading Activity

This table compares the aggregate Proportion of Gains Realized (PGR) to the aggregate Proportion of Losses Realized (PLR), where PGR is the number of realized gains divided by the number of realized gains plus the number of paper (unrealized) gains, and PLR is the number of realized losses divided by the number of realized losses plus the number of paper (unrealized) losses. The data are partitioned into the periods 1987–1990 and 1991–1993 and into the 10 percent of the accounts that trade most frequently and the 90 percent that trade least frequently. For 1987–1990 there are 7,280 realized gains, 28,998 paper gains, 7,253 realized losses, and 50,540 paper losses. For 1991–1993 there are 6,603 realized gains, 50,660 paper gains, 4,677 realized losses, and 59,808 paper losses. For frequent traders there are 10,186 realized gains, 75,182 paper gains, 8,886 realized losses, and 103,096 paper losses. For infrequent traders there are 3,697 realized gains, 4,476 paper gains, 3,042 realized losses, and 7,251 paper losses. The *t*-statistics test the null hypotheses that the differences in proportions are equal to zero assuming that all realized gains, paper gains, realized losses, and paper losses result from independent decisions.

	1987–1990	1991–1993	Frequent Traders	Infrequent Traders
Entire year PLR	0.126	0.072	0.079	0.296
Entire year PGR	0.201	0.115	0.119	0.452
Difference in proportions	–0.075	–0.043	–0.040	–0.156
<i>t</i> -statistic	–30	–25	–29	–22
December PLR	0.143	0.110	0.095	0.379
December PGR	0.129	0.097	0.084	0.309
Difference in proportions	0.014	0.013	0.010	0.070
<i>t</i> -statistic	1.9	2.3	2.3	3.5
Jan.–Nov. PLR	0.123	0.069	0.078	0.282
Jan.–Nov. PGR	0.207	0.117	0.123	0.469
Difference in proportions	–0.084	–0.048	–0.045	–0.187
<i>t</i> -statistic	–32	–27	–31	–25

5.4 Table II: partitions by period and trading activity

Read:

- The result holds separately in 1987–1990 and 1991–1993.
- It also holds for the most active 10% of traders and for the less active 90%.
- The most active 10% of traders account for about 57% of all stock trades.
- In all main partitions, the full-year PGR exceeds PLR by economically large amounts.

Takeaway. The result is not driven by one subperiod or by a small set of unusually active traders.

Table III
Average Returns

This table reports the mean return realized on stocks sold for a gain and on stocks sold for a loss. It also reports mean return that could be realized by stocks that are not sold on days that other stocks in the same portfolio are sold. These stocks are classified as paper gains and paper losses. For all accounts over the entire year, there are 13,883 realized gains, 79,658 paper gains, 11,930 realized losses, and 110,348 paper losses. For all accounts during the month of December, there are 866 realized gains, 7,131 paper gains, 1,555 realized losses, and 10,604 paper losses.

	Jan.–Nov.	December	Entire Year
Return on realized gains	0.275	0.316	0.277
Return on paper gains	0.463	0.500	0.466
Return on realized losses	−0.208	−0.366	−0.228
Return on paper losses	−0.391	−0.417	−0.393

5.5 Table III: average returns on realized and paper positions

	Jan.–Nov.	December	Entire year
Return on realized gains	0.275	0.316	0.277
Return on paper gains	0.463	0.500	0.466
Return on realized losses	−0.208	−0.366	−0.228
Return on paper losses	−0.391	−0.417	−0.393

Read:

- Investors tend to sell smaller gains and hold larger gains.
- Investors tend to sell smaller losses and hold larger losses.
- December realized losses are much larger than losses realized during the rest of the year.

Takeaway. The December behavior looks tax-motivated: investors realize larger losses when the tax deadline approaches.

5.6 Tables IV and V: rebalancing checks

Table IV

PGR and PLR When the Entire Position in a Stock Is Sold

This table compares the aggregate Proportion of Gains Realized (PGR) to the aggregate Proportion of Losses Realized (PLR), where PGR is the number of realized gains divided by the number of realized gains plus the number of paper (unrealized) gains, and PLR is the number of realized losses divided by the number of realized losses plus the number of paper (unrealized) losses. In this table losses and gains are counted only if a portfolio's total position in a stock was sold that day. Paper (unrealized) gains and losses are counted only if the portfolio's total position in another stock held in the portfolio was sold that day. Realized gains, paper gains, losses, and paper losses are aggregated over time (1987–1993) and across all accounts in the dataset. PGR and PLR are reported for the entire year and for December only. For the entire year there are 10,967 realized gains, 36,033 paper gains, 9,476 realized losses, and 51,502 paper losses. For December there are 666 realized gains, 3,440 paper gains, 1,171 realized losses, and 4,759 paper losses. The *t*-statistics test the null hypotheses that the differences in proportions are equal to zero assuming that all realized gains, paper gains, realized losses, and paper losses result from independent decisions.

	Entire Year	December
PLR	0.155	0.197
PGR	0.233	0.162
Difference in proportions	−0.078	0.035
<i>t</i> -statistic	−32	4.6

Table V

PGR and PLR When No New Stock Is Purchased Within Three Weeks of Sale

This table compares the aggregate Proportion of Gains Realized (PGR) to the aggregate Proportion of Losses Realized (PLR), where PGR is the number of realized gains divided by the number of realized gains plus the number of paper (unrealized) gains, and PLR is the number of realized losses divided by the number of realized losses plus the number of paper (unrealized) losses. In this table losses and gains are counted only if a no new purchase was made into a portfolio on the day of the sale or within three weeks following the sale. Paper (unrealized) gains and losses are counted for days on which qualifying sales were made. Realized gains, paper gains, losses, and paper losses are aggregated over time (1987–1993) and across all accounts in the dataset. PGR and PLR are reported for the entire year and for December only. For the entire year there are 8,336 realized gains, 10,240 paper gains, 7,553 realized losses, and 19,370 paper losses. For December there are 590 realized gains, 1,024 paper gains, 1,194 realized losses, and 1,863 paper losses. The *t*-statistics test the null hypotheses that the differences in proportions are equal to zero assuming that all realized gains, paper gains, realized losses, and paper losses result from independent decisions.

	Entire Year	December
PLR	0.281	0.391
PGR	0.449	0.366
Difference in proportions	−0.168	0.015
<i>t</i> -statistic	−36	1.6

Table IV, full-position sales:

- Entire year: PGR = 0.233, PLR = 0.155.
- December: PGR = 0.162, PLR = 0.197.

Table V, no new purchase within three weeks:

- Entire year: PGR = 0.449, PLR = 0.281.
- December: PGR = 0.366, PLR = 0.391.

Read:

- Full-position sales are unlikely to be pure rebalancing trades.
- Sales not followed by new purchases are also less likely to be rebalancing trades.
- In both cases, gains are still realized more often than losses over the full year.

Takeaway. Rebalancing cannot explain the disposition effect.

5.7 Table VI: ex post returns

	Next 84 days	Next 252 days	Next 504 days
Winning stocks sold	0.0047	0.0235	0.0645
Paper losses	-0.0056	-0.0106	0.0287
Difference	0.0103	0.0341	0.0358
<i>p</i> -value	0.002	0.001	0.014

Read:

- Winners sold subsequently outperform paper losers at 84-day, 252-day, and 504-day horizons.
- The one-year gap is about 3.41 percentage points.
- This is inconsistent with the idea that investors are rationally holding losers because those losers are about to outperform.

Takeaway. Investors' reluctance to sell losers is costly ex post.

Table VI
Ex Post Returns

This table compares average returns in excess of the CRSP value-weighted index to stocks that are sold for a profit (winning stocks sold) and to stocks that could be, but are not, sold for a loss (paper losses). Returns are measured over the 84, 252, and 504 trading days subsequent to the sale of a realized winner and subsequent to days on which sales of other stocks take place in the portfolio of a paper loser. *p*-values refer to the frequency with which differences in excess returns over the same periods in the empirical (bootstrapped) distributions exceed the difference in excess returns observed in the data.

	Performance over Next 84 Trading Days	Performance over Next 252 Trading Days	Performance over Next 504 Trading Days
Average excess return on winning stocks sold	0.0047	0.0235	0.0645
Average excess return on paper losses	-0.0056	-0.0106	0.0287
Difference in excess returns (<i>p</i> -values)	0.0103 (0.002)	0.0341 (0.001)	0.0358 (0.014)

Table VII
PGR and PLR Partitioned by Price and Return

This table compares the aggregate Proportion of Gains Realized (PGR) to the aggregate Proportion of Losses Realized (PLR), where PGR is the number of realized gains divided by the number of realized gains plus the number of paper (unrealized) gains, and PLR is the number of realized losses divided by the number of realized losses plus the number of paper (unrealized) losses. The data are partitioned on stock price and on absolute value of the return to date (*R*), for all accounts, 1987–1993, January through November only. The *t*-statistics test the null hypotheses that the differences in proportions are equal to zero assuming that all realized gains, paper gains, realized losses, and paper losses result from independent decisions.

	$ R \leq 0.15$	$0.15 < R \leq 0.30$	$0.30 < R \leq 0.50$	$0.50 \leq R $
Price $\leq \\$10$				
PLR	0.141	0.129	0.109	0.030
PGR	0.267	0.257	0.295	0.282
Difference	-0.126	-0.128	-0.186	-0.252
<i>t</i> -statistic	13.0	10.2	11.7	17.5
\$10 $\leq$ Price $\leq$ \$25				
PLR	0.138	0.105	0.076	0.058
PGR	0.222	0.186	0.172	0.135
Difference	-0.084	-0.081	-0.096	-0.077
<i>t</i> -statistic	16.9	13.1	13.1	11.3
\$25 $\leq$ Price				
PLR	0.125	0.104	0.104	0.049
PGR	0.197	0.126	0.081	0.055
Difference	-0.072	-0.022	0.023	-0.006
<i>t</i> -statistic	19.4	4.5	-3.2	0.61

5.8 Table VII: price and return partitions

Read:

- The paper partitions observations by stock price and by the absolute magnitude of the gain or loss.
- In 14 of 15 price–return cells, PGR exceeds PLR.
- The difference is statistically significant in 13 of the 15 cells.

Takeaway. Higher transaction costs for low-priced stocks do not explain why investors hold losers. Even within similar price and return-magnitude groups, investors prefer to realize winners.

5.9 Additional-purchases evidence

Read:

- Investors are more likely to buy more shares of losing positions than winning positions.
- The key rates are $PLPA = 0.135$ and $PGPA = 0.094$.
- The difference is large and statistically strong under the paper’s baseline independence assumption.

Takeaway. Investors do not merely avoid trading loser stocks because of transaction costs. They are willing to trade losers when the trade increases the position. This supports the behavioral interpretation.

5.10 Discussion calculation: economic cost

Read:

- The paper gives a simple calculation for an investor choosing whether to sell a \$1,000 winner or a \$1,000 loser.
- Selling the winner realizes a taxable gain; selling the loser realizes a tax loss.
- Using the paper’s average realized winner, average paper loser, a 15% tax rate, and an 8% return assumption, selling the loser rather than the winner raises the next-year return by about **4.4%** on the \$1,000 basis.

Takeaway. The disposition effect is not just statistically significant. It can be economically meaningful for individual investors, especially in taxable accounts.

6 Short summary

- **Method.** Compare the rate at which investors realize gains to the rate at which they realize losses, using actual brokerage-account trading records and opportunity-adjusted denominators.
- **Data.** 10,000 discount brokerage accounts, 1987–1993, matched to CRSP stock data.
- **Core measure.** $PGR = RG/(RG + PG)$ and $PLR = RL/(RL + PL)$.
- **Main result.** Investors realize gains much more readily than losses: full-year PGR is 0.148 versus PLR of 0.098.

- **Tax result.** In December the relation reverses, with PLR exceeding PGR, consistent with tax-loss selling.
- **Robustness.** The disposition effect survives account-level tests, share- and dollar-based measures, subgroup partitions, rebalancing controls, price controls, and ex post performance tests.
- **Main interpretation.** Investor behavior is consistent with reference-dependent preferences and reluctance to realize losses; it is also consistent with mistaken mean-reversion beliefs, but those beliefs are not justified by subsequent returns.

7 Conclusion

7.1 Core conclusion

- Individual investors show a strong disposition effect.
- They sell profitable stock positions faster than unprofitable ones.
- The only major exception is December, when tax-loss selling makes investors more willing to realize losses.

7.2 Economic intuition

- A gain is psychologically easy to close because it confirms success.
- A loss is psychologically painful to close because it forces the investor to admit failure.
- As a result, investors keep losers alive, sometimes even buying more, while selling winners that continue to perform well.

7.3 Takeaway

Odean's paper is a foundational field-data test of the disposition effect. Its central lesson is simple: individual investors do not treat realized and unrealized gains and losses symmetrically. They realize gains too readily, defer losses too long, and in taxable accounts this behavior lowers after-tax returns.